

CCNA LAB: Inter-VLAN Routing -Lab 10

1. Lab Objective

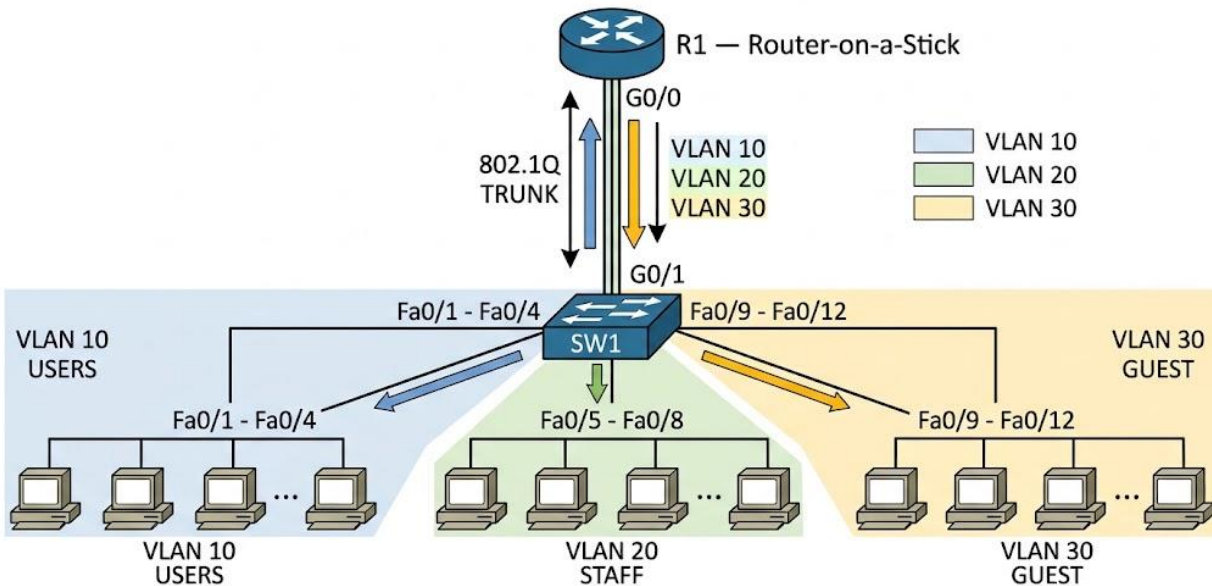
- ❖ Create three VLANs on a Cisco switch.
- ❖ Assign access ports to the correct VLANs.
- ❖ Configure an 802.1Q trunk between the switch and router.
- ❖ Configure Inter-VLAN Routing using Router-on-a-Stick.

Detailed Explanation / Configuration

By default, hosts within the same VLAN can communicate with each other since they are part of the same broadcast domain. When a host tries to communicate with the other host, the remote system MAC address should be discovered using ARP broadcast. The ARP broadcast can be spread out within a broadcast domain only. Let's recall that each VLAN is considered as an individual broadcast domain.

2. Topology

Router-on-a-Stick Configuration Diagram



3. IP Addressing Plan

VLAN	Name	Network	Gateway	DHCP Range	Access Ports
10	USERS	192.168.10.0/24	192.168.10.1	192.168.10.11–254	Fa0/1–4
20	STAFF	192.168.20.0/24	192.168.20.1	192.168.20.11–254	Fa0/5–8
30	GUEST	192.168.30.0/24	192.168.30.1	192.168.30.11–254	Fa0/9–12

4. Device Configuration

R1 — Router Configuration

enable

configure terminal

```
hostname R1
```

```
Physical interface connected to SW1
```

```
interface GigabitEthernet0/0
```

```
no shutdown
```

```
exit
```

In the above configuration, note that we used “no shutdown” command under the physical interface while the IP addresses are configured under the sub-interfaces created for each VLAN. The number of sub-interfaces is equal to the number of VLANs as required for the inter-VLAN routing.

Note: make sure that the default gateways are configured on each host with the configured IP addresses under the sub-interface for each VLAN.

VLAN 10

```
interface GigabitEthernet0/0.10
```

```
encapsulation dot1Q 10
```

```
ip address 192.168.10.1 255.255.255.0
```

```
description VLAN10-USERS
```

```
exit
```

VLAN 20

```
interface GigabitEthernet0/0.20
```

```
encapsulation dot1Q 20
```

```
ip address 192.168.20.1 255.255.255.0
```

```
description VLAN20-STAFF
```

```
exit
```

VLAN 30

```
interface GigabitEthernet0/0.30
```

```
encapsulation dot1Q 30
ip address 192.168.30.1 255.255.255.0
description VLAN30-GUEST
exit
end
write memory
```

SW1 — Switch Configuration

```
enable
configure terminal
hostname SW1

Create VLANs

vlan 10
name USERS
exit
```

```
vlan 20
name STAFF
exit
```

```
vlan 30
name GUEST
exit
```

```
VLAN 10 access ports

interface range FastEthernet0/1 - 4
switchport mode access
```

```
switchport access vlan 10
description VLAN10-USERS
exit
```

VLAN 20 access ports

```
interface range FastEthernet0/5 - 8
switchport mode access
switchport access vlan 20
description VLAN20-STAFF
exit
```

VLAN 30 access ports

```
interface range FastEthernet0/9 - 12
switchport mode access
switchport access vlan 30
description VLAN30-GUEST
exit
```

Trunk to R1

```
interface GigabitEthernet0/1
switchport mode trunk
switchport trunk allowed vlan 10,20,30
description TRUNK-TO-R1
no shutdown
exit
end
write memory
```

CCNA LAB: Inter-VLAN Routing with DHCP-LAB11

VLANs • Router-on-a-Stick • DHCP • Trunking • End-to-End Testing

R1 — Router Configuration

Exclude gateway and first 10 addresses

```
ip dhcp excluded-address 192.168.10.1 192.168.10.10
```

```
ip dhcp excluded-address 192.168.20.1 192.168.20.10
```

```
ip dhcp excluded-address 192.168.30.1 192.168.30.10
```

DHCP VLAN 10

```
ip dhcp pool VLAN10-USERS
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1
dns-server 8.8.8.8
domain-name users.local
exit
```

DHCP VLAN 20

```
ip dhcp pool VLAN20-STAFF
network 192.168.20.0 255.255.255.0
default-router 192.168.20.1
dns-server 8.8.8.8
domain-name staff.local
exit
```

DHCP VLAN 30

```
ip dhcp pool VLAN30-GUEST
network 192.168.30.0 255.255.255.0
default-router 192.168.30.1
dns-server 8.8.8.8
domain-name guest.local
exit
end
write memory
```

Verification Commands

SW1# show vlan brief

SW1# show interfaces trunk

SW1# show interfaces gigabitEthernet0/1 switchport

R1# show ip interface brief

R1# show ip route

R1# show ip dhcp pool

R1# show ip dhcp binding